

ENDOCRINE SYSTEM

NCLEX STUDY GUIDE



Organized for study and printing | 16 content pages

ENDOCRINE SYSTEM

Pituitary gland

ANTERIOR PITUITARY	POSTERIOR PITUITARY
1. Growth hormone ✓ Dwarfism (varones) Enanism ✓ Acromegaly. 2. Thyrotropin (TSH) ✓ Hyperthyroidism ✓ Hypothyroidism 3. Adrenocorticotrophic (ACTH) ✓ Addison ✓ Cushing 4. FSH or follicle stimulant (menstruation) 5. Luteinizing (LH)	- Oxytocin - Vasopressin ADH ✓ SIADH ✓ Diabetes Insipidus (DI)

Sheehan 's syndrome = Hypopituitarism = todo ↓

- Woman who loses a large amount of blood in child bird or
- A severe decrease BP after labor can deprive the amount of oxygen of the pituitary gland.

Assessment:

- Inability to breast feeding
- Amenorrhea (no menstruation)
- Inability to regrow shave public hair
- Slow mental function (hypotiroidism)
- Weight gain
- Low BP
- Low B6
- Cardiac dysrhythmias

Treatment

- Lifelong hormone replacement

DWARFISM (delicate features)	ACROMEGALY (Coarse features)
- Decreased GH - Assessment Delicate features - height below normal - Body proportion are normal - Bone and tooth development delayed - Sexual maturity and organs delayed - Fine skin	Cause= tumor in the hypophysis • After 25 years old • Increased GH <u>Assessment</u> 1. Thickened lips 2. Increased head size 3. Lower jaw protrusion 4. Hyperglycemia

<u>Treatment</u> 1. Somatotropin – (6H) GH replacement therapy until maturity (18 – 20 years old). 2. Gonadotropin chorionic 3. Emotional support. 4. Testosterone replacement	5. Enlarged feet and hands 6. Joint pain 7. Deep voice 8. Coordination poor 9. Visual field changes 10. Liver, heart, lungs enlargement <u>Diagnostic</u> = Increased GH <u>Treatment = OJO</u> Bromocriptine: is used to for neuroleptic malignant, hyperthermia malignant and Parkinson. Octreotide = Somatostatin (stop grow) <u>Surgical Intervention (remove tumor)</u> - Trans nasal Hypophysectomy <u>Post – op - Intervention:</u> 1. HOB = 30 (head at the midline) bc expected ↑ ICP 2. Mustache dressing and nasal package will be placed on the nose. (Halo sign) 3. Teaching = Client must be breath through the mouth for 3-4 days after surgery. 4. Avoid activities that ↑ ICP <u>Complication</u> 1-Diabetes Insipidus (too much urine)> 150 ml (normal (30-50 in 1h)) 2-Cerebro-Spinal fluid (leakage) = Call Dr ♦ Halo sign in mustache dressing ♦ Positive to glucose liquid = Bedrest 30 3-Meningitis = neck stiffness, headache and fever 4-Expected ↓ of sense of smell 4-6 weeks after 5-Avoid activities that ↑ ICP *no cepillarse los dientes 2 weeks after, solo enjuague y hilo dental.
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Nota: Acromegalia = crecen todos los órganos.

Diabetes insipidus	Syndrome inappropriate ADH
Cause = Brain 1. Hypophisectomy 2. Head trauma 3. Meningitis 4. Brain tumor 5. Lithium Therapy Assessment 1. Increased urinary output (polyuria) 2. Increased thirst 3. Polydipsia Treatment 1. IV fluid = 0.9 % NaCl - Hypotonic solution choice (0.45 or ½) 2. Desmopressin (Oral or nasal solution) Goal: ↓ Urinary output ↓ thirst Specific gravity ↑ (1005 -1030) LAB = - Urinary SG = less than 1005 - Plasma urinary ↓ - Plasma osmolality ↑ - Na ↑ - BUN ↑ - Hematocrit ↑ - Capillary refill > 3 Complication 1. SIADH - Stop urinary output - Specific gravity = 1030 - LOC alteration, confusion 2. ↑ BP, MI, Cardiac dysrhythmias	Cause = ↑ADH - Lung: 1. Small cell lung cancer 2. Pneumonia 3. Mechanical Ventilation 4. TB Assesment 1. Sign of increased fluid overload = CHF. ✓ SOB ✓ Crackle ✓ Cough 2. Decrease urinary Output 3. Increased BP + tachycardia with bounding pulse 4. Lethargy, decreased DTR 5. Brain edema, LOC alteration, confusion, seizure LAB = En la orina todo alto y en el cuerpo todo bajo. ✓ USG > 1030 ✓ Plasma urinary ↑ ✓ Plasma osmolality ↓ ✓ Na ↓ ✓ BUN ↓ ✓ Hematocrit ↓ Treatment 1. Water restriction (500-600ml in 24 hours. 2. Diuretic (furosemide) 3. IV fluid 3% NaCl 4. Demeclocycline (antibiotic + antidiuretic propertie- primo de la tetraciclina) lab check = K (while administering)= Side effect ✓ Teeth discoloration ✓ Photosensitivity ✓ No in children less than 9 years. ✓ No with milk Goal ✓ Increase urinary output ✓ Decreased urinary specific gravity ✓ Improve LOC + diabetes Insipidus

Thyroid

Hypothyroidism (tiene frio)	Hyperthyroidism (tiene calor)
✓ Myxedema. ✓ Hashimoto thyroiditis (typo of hypothyroidism) • Decreased activity level Diagnostic T3, T4 ↓, TSH ↑ <u>Assessment</u> 1. Increased sensitive to cold 2. Weight gain (basal metabolism ↓) 3. Dry skin 4. Decreased stimuli perception 5. Respiratory difficulty 6. Constipation 7. Alopecia 8. Bradycardia 9. Decreased sweating 10. Reproductive problems 11. Low cardiac output (hypotension, fatigue) 12. Peripheral edema 13. Slow mental process (teaching; speech and writing) 14. Hypercholesterolemia <u>Intervention hypothyroidism</u> 1. Medical Alert Bracelet 2. No OTC medication 3. Diet: high fiber, low carbohydrates. 4. Frequent rest periods between activities. 5. Teaching speech and writing. 6. Maintain temperature 75° F Medication Levothyroxine = Take in the morning before 7:00 am with an empty stomach + plenty fluid. Goal: normal functioning)	Gravis disease (type of hyperthyroidism) =hyperthyroidism + exophthalmos Diagnostic: Increased T3, T4 ↑, TSH ↓ ✓ Uptake of iodine Radioactive iodine Less than 35 = OK More than 35 = hyperthyroidism <u>Assessment</u> 1. Hyperactivity 2. Increased sensitive to heat 3. Increased temperature 4. Rest and sleep deprivation 5. Increased stimuli perception 6. Weight loss 7. Diarrhea 8. Diaphoresis 9. Tachycardia 10. Increased BP 11. Cardiac dysthymia 12. Fine soft hair 13. Exophthalmos 14. Photophobia 15. Chest pain 16. Mood swings. <u>GOITER classification</u> Grades ✓ O – No palpable, no visible (ultrasound) ✓ 1- Mass no visible with neck in normal position, but can be palpable when Pt swallowing (se palpa por detras)

<u>Side effects</u> ✓ Hyperthyroidism. -<u>Thyroid storm = complication of levothyroxine</u> ✓ Fever ↑1° F temp. ✓ ↑ Tachycardia <u>Complication of hypothyroidism</u> ✓ Mixedema Coma <u>Triggers factors.</u> ✓ Acute illness ✓ Surgery ✓ Chemotherapy ✓ Stop thyroid replacement hormone (levothyroxine) ✓ Use of sedative and opioids <u>Assessment</u> ✓ LOC alteration, confusion, lethargy, coma. ✓ Respiratory failure (Respiratory Acidosis). ✓ Severe Hypotension ✓ Severe Hyponatremia ✓ Severe Hypothermia ✓ Severe Hypoglycemia. <u>Intervention</u> 1. Patency airway 2. IV fluids 3% NaCl (hypertonic) 3. Levothyroxine IV 4. Glucose IV 5. Steroids 6. Monitor temperature every hour ✓ Cover with warm blanket 7. Change position every 2 hours 8. Aspiration precaution. Suction equipment at bed side. Note: resume	✓ 2- Mass is visible as swelling when the neck is in the normal position, goiter is easily palpable and is usually asymmetric <u>Intervention</u> 1. ↓ Environmental stimulation. – restrict visitors. 2. Ensure that client has a fresh pitcher of ice water at bed side 3. Change lines frequently, cold water shower <u>Medications</u> 1- Methimazole (contraindicate 1st trimester pregnancy) 2- Propylthiouracil PTU <u>S/E for both:</u> -↓ WBC = Risk for infection – if fever or sore throat = call Dr. Note: Med that ↓ WBC = Clozapine Clozaril, carbamazepine, and hyperthyroidism med. (contraindicated in immunocompromised) -Liver toxicity -Hypothyroidism, (Myxedema coma) If bradycardia call Dr. 3- Lugol Solution (oral liquid) -Use for shorth term (1-2weeks) -1 week before thyroid surgery or -to treat thyroid storm S/E Teeth discoloration = use straw (same tetracycline) Metallic taste Skin rash
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Hypothyroidism = Myxedema Hypothyroidism + Levothyroxine = Thyroid storm Hyperthyroidism = Thyroid storm Hyperthyroidism + PTU and Methimazole = Myxedema Coma	4- Ablation Therapy - Radioactive Iodine -RAI 131 - No in pregnant woman - No in childbearing age - Fasting overnight 8 – 12 hrs. before. - Urine, saliva, semen, vomit = radioactive for 1 week after (5-7 days) - Flushing toilet twice for each voiding. - Wash your hand with soap and water. - Use gloves is any contact with secretions - During the first 6 – 8 weeks after radiation therapy. Continuous with antithyroid medication using methimazole or PTU. - After that = levothyroxine lifelong (for hypothyroidism) 5- Thyroid Surgery (thyroidectomy) (partial or total) OJO Post-surgical intervention and complication: ✓ Position semi fowler to fowler (30-45) HOB ✓ Avoid neck hyperextension ✓ Use a sandbag to support the neck or put one hand behind the neck to support when walking. ✓ Use “Jaw thrust” Maneuver to airway assessment. ✓ Hoarseness is expected during the first week after surgery, if worsening or persisting more than 1 week then call Dr. <u>Complication of the thyroidectomy.</u> - 1. Hypocalcemia = Emergency. ○ Paresthesia's (numbness tingling sensation on mouth or fingers) ○ Chvostek. ○ Trousseau 2. Laryngeal spasm (stridor) Calcium Gluconate & Tracheostomy Set at bed side. 3. Hemorrhage
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	• Check behind the neck • If respiratory distress call Dr 4. Thyroid storm Triggers • Stress • Thyroid Manipulation • Acute Illness • Surgery • Thyroid Surgery • Stop antithyroid meds. Assessment • Fever or ↑1 F temperature • Tachycardia (atrial Fib) • Anxiety, • tremors • N/V • Delirium, • Agitation • Zeusure. Treatment • ICU admission • Colling blanket • Lugol solution • Antipyretic • Propranolol (no in Asthma) • PTU or methimazole IV •
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Pheochromocytoma (suprarenal Gland tumor)

- Bening tumor of the adrenal gland.
- Do not palpate
- Hypertension crisis (no MAOIS)

Assessment

1. Paroximal or sustained Hypertension (high BP)
2. Trobbing Headache.
3. Tachycardia and palpitation.
4. Flushing
5. Profuse diaphoresis- frequent bath with warm water
6. Heat intolerant

7. Weight loss
8. Tremor
9. Hyperglycemia

Diagnostic

- Vanil mandelic Acid. (VMA) = 24 hours urinary collection
- Avoid 72h before the test.
 - o Coffee
 - o Bananas
 - o Vanilla
 - o Chocolate
- > 5/24h = positive for Tumor.

Treatment

1. Check for BP to prevent complications. (Risk for stroke or MI)
2. Medication = B-Blockers (LOL) to control BP
3. During crisis use **Phentolamine** (vasodilator) (choice) sino hay este, dar:
 - o **Sodium nitroprusiate** (es photosensitivity, use in ICU, blood pressure med potente, la luz lo inactiva, si se pone green or red tienes que botarlo, para ponerlo el Pt tiene que estar en cama y checarle la BP porque la baja mucho)
 - o Hydralazine (use in pregnancy)
 - o Nifedipine (choice for Autonomic dysreflexia)
4. Decrease environmental stimulation
5. Restrict visitor,
6. Bed rest
7. Do not palpate Abd sign bc increase BP
8. Diet: Low Na, avoid stimulant
 - a. High in calories, vitamins and minerals.
9. Frequently bath with warm water
10. **Adrenalectomy**

Nota importante: **Phentolamine** (vasodilator): use for extravasation of vasoconstrictor
Ex; epinephrine or dopamine, el antidote es phentolamine que se Administra subcutaneous.

Note: Wilms Tumor: No biopsy, No palpate bc metastases.

Conn's Disease

= Primary hyperaldosteronism

Agua y sodio para mi y potasio para el toilet

No with digoxin bc digoxin is toxic when K is low

No with steroids bc decrease K

No with licorice (herb) bc decrease K or Ginseng

Lithium si se puede dar bc Na esta alto, y el es toxico cuando el Na es bajo.

Assessment

1. Increased Na
2. Fluid retention
3. Hypokalemia (U wave)
4. Cardiac dysrhythmias
5. Paresthesia = Tetany
6. Visual change
7. Glucose Intolerance

Treatment

1. Administer K supplement
2. K sparing diuretics
 - Spironolactone
 - Triamterene
 - Amyloride
3. Steroids
4. Glucocorticoid

ADDISON (adipositas)(Stress)	CUSHING
Decrease Steroids - ↓ Cortisol - ↓ Aldosterone Assessment - K – Ca ↑ - Skin hyperpigmentation or brown skin discoloration. - Tachycardia - ↓ Weight - Hypotension - ↓ Na - Lethargy, Fatigue - Muscles weakness - ↓ Blood glucose - Menstrual change - Impotence in male Diagnostic ACTH Stimulation Test = (Confirmatory) - 17 hydroxy corticosteroids - 17 Ketosteroid. Treatment. 1. Avoid triggers to prevent Addisonian crisis. • Stress • Inadequate steroid replacement • Infection • Surgery Note: Stop suddenly, steroids also can cause Addisonian crisis, that's why needed to increase steroid in a surgery. Diet - ↑ Na and Low K - Table salt ↑ (same in Lithium) - Salt substitute ↓ bc is K - High protein	Too much steroids - Increased cortisol - Increased aldosterone Assessment - Ca -- K ↓ - Moon Face - "Buffalo Lump" (back) - ↑ weight (fluid retention) - Gynecomastia - Hirsutism (face and body hair) - Muscle wasting (buttocks, extremities) - Abdominal stretch - Easy bruising (Petechiae) - Healing retardation. - Dry skin - ACNE - Osteoporosis (safety precautions) med: Alendronate - Cataracts - Glaucoma - Immunocompromise - High Risk for infection - ↑ BP (hypertension) - ↑ BG (hyperglycemia) - ↑ Na (hypernatremia) Diagnostic Steroids ↑ Intervention 1. Diet • ↑ high protein • ↓ Low carbohydrate • ↓ Low Na • ↓ Calorie • ↑ K, ↑ Ca • Fluid Restriction

- ↑ Carbohydrates - Wear a medical alert bracelet - Lifelong steroid replacement - Fludrocortisone (Choice) Oral. Sino hay pueden dar entonces : Prednisone Complication - Addisonian Crisis Assessment • Hypotension + Tachycardia • Epigastric pain • Severe headache • Confusion • Hyponatremia severe • Hypoglycemia severe Treatment 1. IV fluids (Hypertonic solution 3%NaCl) 2. Hydrocortisone (IV) 3. Fludrocortisone PO (pastillas only) Note: Isotonic Solution: misma cantidad de agua que de Sodio • NaCl 0.9% • Ringer lactate • 5% Dext in water • 5% Dext in 0.225 NaCl Hypotonic = un # < 1 Ex: ¼ NaCl, 0.225NaCl = H₂O Ex: near drowning in the Pacific Ocean. Hypertonic = un # > 1 = Na Ex: near drowning in fresh water.	2. Assist with ambulation - Elimination of environmental hazards to prevent Osteoporosis. Medication 1. Aminoglutethimide 2. Hypophysectomy or Adrenalectomy Emotional support The most physical changes are reversible with treat. Post adrenalectomy care - Lifelong Steroids - High risk for infection - Risk for respiratory acidosis) (Acostarse sobre la herida 2-3 veces al dia para prevenir respiratory acidosis) Note: Cholecystectomy and nephrectomy or adrenalectomy cause respiratory acidosis, 2-3 times per day acuestate sobre la herida para prevenirlo.
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Diabetes Mellitus

- Chronic disorder of impaired carbohydrate, protein and lipids metabolism caused by deficiency of insulin or by increase peripheral resistance to insulin.

Type I Diabetes	Type II Diabetes
- Absolute Insulin deficiency (B cell destruction) - More frequently in childhood before 30 years old. - Insulin Dependent	- Relative lack of insulin or increase peripheral resistance. - 80% = Obesity (abdominal) - Peak = 50 yrs.

Note: difference between diabetes mellitus and insipidus is polyphagia, that is for mellitus.

Metabolic syndrome

Acanthosis nigricans = Predisposition of 90% diabetes

1. Bbd Obesity
 - Waist circumference + 40 inch for men, + 35 inches for women
 - Apple shape
2. Hyperglycemia

- Fasting glucose 70-110 = normal
- 110-125 Metabolic syndrome
- >126 = Diabetes

A1C = less than 5.6 = normal

- More than 6.5 = Diabetes
- 5.6-6.4 = Metabolic syndrome

7 for control value

Less than 7 se porto bien

More than 7 = Bad control

3. Systolic BP between 130-140
 - Diastolic BP 85-90

More than 140-90 = Hypertension.

4. Hyperlipidemia
 - ◆ Triglycerides > 150
 - ◆ HDL cholesterol >50 = ok
 - ◆ Less than 40 in Men and 50 in Women = mal

Assessment for Diabetes Mellitus.

1. Polyuria (increased urine)
2. Polydipsia (increased thirst)
3. Polyphagia (increased appetite)
4. Decreased weight
5. Hyperglycemia
6. Healing retardation
7. High risk for vaginal Infection (Fungal)
8. Blurred Vision

Complication

Short Term	Long Term
1. Hypoglycemia 2. Diabetes Ketoacidosis (Type 1) 3. Hyperosmolar non ketonic stage (Type II) (coma hyperosmolar)	1. Chronic Complication ◇ Diabetes Retinopathy = legal blindness 20/200 ◇ Diabetes Nephropathy Initially proteinuria = chronic kidney diseases Lasting, = Renal insufficiency ◇ Diabetes Polyneuropathy = Diabetic foot ◇ Reproductive: - Impotence - Menstrual change - Dry vaginal

Hypoglycemia

- Blood glucose level falls below 70 mg/dl
 - Mild less than 70
 - Moderate less than 40
 - Severe less than 20

Assessment

- Hunger
- Palpitation/ tachycardia

- Drowsiness
- Nervousness
- “Quivering in abdomen”
- Jitteriness (infant) (newborn)
- Doble visión
- Headache
- Sweating
- Slurred speech
- Confusion
- Difficulty to arouse
- Disorientation
- Seizure
- Loss of consciousness

Treatment

- Carbohydrate replacement 15 – 20g of glucose
 - ✓ ½ cup (4 – oz of fruit juice)
 - ✓ ½ cup of regular soft drink
 - ✓ 8 oz of skinny milk.
 - ✓ 6 – 10 candies or live savers
 - ✓ 4 cubes of sugar
 - ✓ 4 teaspoons of sugar
 - ✓ 3 graham crackers
 - ✓ 6 saltines crackers

Le doy una de ellos nada mas and

- 15 minutes after check blood glucose,
- If continue hypoglycemic = repeat one more
- 15 after checking blood glucose again.

If hypoglycemia administers small snack of carbohydrates + protein.

- If 15 min after continuing hypoglycemia after small snack = **Severe hypoglycemia.**
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<u>Consciousness</u>	<u>Unconsciousness</u>
- Dextrose	- Glucagon - Repeat doses after 10 mins.

Nota: Tto de elección para la diabetes + HTN = prietas. But if already renal failure = NO PRILES (bc increased K)

Diabetes ketoacidosis (type of metabolic acidosis) Complication type I		Hyperosmolar (Hyperglycemic nonketotic) (HTTS) Type II
Onset and Precipitating Factors	Sudden - Infection - Inadequate Insulin Dose - Surgery	Gradual • Poor fluid intake • Infection
Assessment		

- Kussmaul respiration - Fruity breath Odor - Epigastric Pain - Sign of hyperglycemia ❖ Polyuria, polydipsia, polyphagia and loss weigh - Dehydration - Tachycardia - Dry and warm skin	- Dehydration (more severe) - LOC Alteration, confusion, coma
Labs	
- Serum glucose >300 – 600 (hyperglycemia) - Plasma Osmolarity variable - Urine Ketones- positive - BUN increased > 20 - Hematocrit increased 35% - K increased with acidosis - PH Acid↓ - HCO₃ ↓	- Serum glucose > 600 - Plasma Osmolality > 350 (severe dehydration). - Urine ketones – negative - K = increase - BUN Increased - Hematocrit Increased - Do not administer Bicarbonate
Intervention (always cardia monitor)	
1. IV fluids NaCl 0.9% or 0.45% NaCl (hypotonic) until blood glucose matches 250 to 300mg. Add dextrose to IV fluid to prevent rebound Hypoglycemia 2. Administer insulin (R) IV (regular only) = IV bolus 5 – 10 units initially then continuous with infusion in IV fluid. 3. Add K to the bag when K level = 3.5 to prevent rebound hypokalemia.	

Diabetic Foot Care

1. Inspect your feet daily, especially between toes.
2. Wash your foot daily with lukewarm water and soap, then dry.
3. Apply moisturizing cream to your feet after bathing. **Avoid between your toes.**
4. Change into clean cotton socks daily.
5. Do not wear the same pair of shoes 2 days in a row, and wear leather or cloth, no sandals, no open toes shoes.
6. Purchase your shoes later in the day.
7. Trim your nails straight across with a clipper. (Never curve, never soak your feet.)
8. Use a mirror to see your feet If nobody can check them for you
9. Blister, corner, never treat at home.
10. Do not step into bathtub without checking the water temperature. Optimal temp = 95F, Normal temp. = 110F.
11. Do. Not walk barefooted
12. Avoid smoking.
13. No hearing pads

Oral Hypoglycemic (only for type 2 diabetes)

Biguanides Metformin	Sulfonylureas No with Ginseng Gli- at the beginning	Thiazolidinedione Tazones = CHF	Meglitinide Glinide at the end	DPP-4 Inhibitors Gliptin at the end	α Glucosidase Inhibitors
Metformin <u>Mechanism action</u> Intestinal glucose transport. Do not cause Hypoglycemia S/E Diarrhea/nausea - ↓ weight - More lethal Lactic Acidosis - Check renal function before *Withhold metformin 24 hours before and 48 hours after any Dye exam.	Glyburide Glipizide Glibenclamide Glimepiride <u>Mechanism action</u> - Lower fasting plasma, blood glucose by stimulate insulin production from β Cell. -Cause hypoglycemics S/E - No in Pts allergic to sulfa. - No with ginseng - No with alcohol To prevent disulfiram reaction. - weight gain	Pioglitazone Rosiglitazone <u>Mechanism action</u> - ↓ liver glucose production reducing fasting plasma and improve insulin receptors. S/E CHF Contraindicated Fluid overload SIADH Note: Tazon lleno Do not cause hypoglycemia but CHF symptoms	Netaglidine Repaglinide <u>Mechanism Action</u> - Same to sulfonylurea) -Cause Hypoglycemia	Saxagliptin Sitagliptin Linagliptin <u>Mechanism Action</u> - Same to sulfonylurea - Delating gastric Empty. Cause Hypoglycemia S/E Pancreatitis Liver toxicity	Acarbose Miglitol <u>Mechanism action</u> - Reduce the rate digestive of stretches. - Delated absorption of carbohydrate from small intestine -Do not cause Hypoglycemia - Give with the first bite of each meal S/E GI Disturbances - Abdominal discomfort - Diarrhea - vomiting - indigestion Contraindicated - In inflammatorily bowel disease (Chron Disease and colitis) - No if bowel obstruction is suspect

Insulin

Injection Site

1. Abdomen
2. Posterior Arm
3. Anterior Thigh
4. Hips

U – 100 syringes = 100 U in 1ml

Administration

Do not massage

Do not aspirate

- ✓ $\angle 45^\circ$ or 90 if BMI >25 = Overweight or obese
- ✓ Needle length = 0,5 – 0,6 – 5/8 inc.
- ✓ Needle gauge = 26 - 27

Note: 18=blood transfusion

22= IV regular

To prevent Lipodystrophy = Systematic Rotation

- Do not use the same site for more than 2 weeks period.
- ½ inch apart within anatomical area.
- Do not Heat, no exercises, no massage after insulin administration.

Storing Insulin

- Avoid exposure to extreme temperature.
- Should not be frozen or kept in direct sunlight.
- Can be refrigerated. Remove 1-2 hours before administered
- Administer at room temperature
- Never freezer
- The vial of Insulin will be used up in 1 month.
- During Illness = increased dose

Types of Insulin

1.Rapid Acting	2.Short Acting- Regular Only use in Emergency	3.Intermediate Acting Isophane NPH (cloudy)	4.Long Acting Do not Mix
Lispro = Humalog Aspart = Novolog Glulisine = Apidra Luis Alberto Garcia	Novolin (R) Relion (R) Humulin (R) Safe in Pregnant R= Peak before 12:00	Novolin (N) Relion (N) Humulin (N)	Lantus Gargling
ONSET- PEAK DURATION			
Onset = 0.25min Peak = 1-3 hrs. Before 10:00 Duration = 3 -5 hrs breakfast, at bed side or ready, If not in the room do not administer	Onset = 0.50 hrs Peak = 2 – 4 hrs Before Noon = 10:00am – 12:00pm Duration = 5-7 h Can be mix Note: Peak = More risk for hypoglycemia	Onset = 1.5 hr Peak = 4 – 12 hrs Between 2:00 – 8:00pm Duration = 16 hrs Cloudy can mix with regular	- No peak -No Mix in the same syringe. -Same to basal insulin, administer daily or twice daily. Duration = 24 hrs

Mix Insulin

NR (Nancy Rodriguez)	RN
Air	Withdraw

1. Wash Hands
2. Gently Rotate NPH(cloudy) insulin bottle

- ✓ Do not shake = Swirl
- 3. Wipe tops of insulin vial with alcohol.
- 4. Draw back amount of air into the syringe that equal total dose
- 5. Inject air equal to NPH dose into NPH vial.
- 6. Remove Syringe.
- 7. Inject air equal to regular dose into regular vial.
- 8. Invert regular insulin bottle and withdrawal regular insulin dose.
- 9. Carefully withdrawal NPH dose.

Nota: **Exenatide (Byetta)** = Used for Type II diabetes (Subcutaneous), because it is not an insulin. Follow up = a Type I Patient wants to get it

Insulin complication

1. Lipodystrophy
 2. Down Phenomenon
 3. Somogyi Phenomenon
- Down Phenomenon
 - Hyperglycemia (Pre- Breakfast)
 - Early in the morning 5:00 am 7:00am
 - Check at 2:00 – 3:00
 - If
 - Normal Glycemia
 - Or Hyperglycemia
 - Tto: Decrease Snack
 - At bet time
 - Increased insulin dose
 - Somogyi Phenomenon
 - Hypoglycemia
 - Tto: Give snack at bet time
 - ↓ Insulin dose NPH 4-10 pm

Insulin Pumps

1. Need check blood glucose apart.
2. Catheter Change every 2 – 3 days
3. Abdomen
4. Deliver a Continuous basal rate of insulin before each meal.